WORKED SOLUTION

STEP I 2019, Question 6

Pure • Inequalities and Bounds

STEP I 2019, Question 6

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

Each equation looks hopeless to solve directly, but it can be split as "something = something else" where the left-hand side can never be smaller than some number and the right-hand side can never be bigger than the same number. Then the equation holds only when *both* sides sit exactly at that number, and each of those conditions is easy to solve. The early parts of each section are there to build the two bounds.

Solution

Part (i)

Completing the square,

$$\begin{aligned} 9x^2 - 12x\cos\theta + 4 &= (3x - 2\cos\theta)^2 - 4\cos^2\theta + 4 \\ &= (3x - 2\cos\theta)^2 + 4\sin^2\theta. \end{aligned}$$

A square is never negative, so the minimum value is $4\sin^2\theta$, attained exactly when $x = \frac{2}{3}\cos\theta$.

Treating the quartic as a quadratic in x^2,

$$12x^2\sin\theta - 9x^4 = -(3x^2 - 2\sin\theta)^2 + 4\sin^2\theta.$$

So $12x^2\sin\theta - 9x^4 \leq 4\sin^2\theta$, with equality exactly when $x^2 = \frac{2}{3}\sin\theta$. Since $0 < \theta < \pi$ we have $\sin\theta > 0$, so this value of x^2 really is achieved, and the maximum value is $4\sin^2\theta$.

Now split the x^2 term of the given equation as $9x^2 - 12x^2\sin\theta$ and rearrange:

$$\underbrace{9x^2 - 12x\cos\theta + 4}_{\geq 4\sin^2\theta} = \underbrace{12x^2\sin\theta - 9x^4}_{\leq 4\sin^2\theta}.$$

Ask yourself How can a quantity that is at least $4\sin^2\theta$ equal one that is at most $4\sin^2\theta$?

Only if both equal $4\sin^2\theta$. So the equation holds if and only if *both*

$$x = \tfrac{2}{3}\cos\theta \qquad \text{and} \qquad x^2 = \tfrac{2}{3}\sin\theta.$$

Eliminating x: $\frac{4}{9}\cos^2\theta = \frac{2}{3}\sin\theta$, so $2\cos^2\theta = 3\sin\theta$, i.e. $2 - 2\sin^2\theta = 3\sin\theta$. Hence

$$2\sin^2\theta + 3\sin\theta - 2 = 0 \quad \Longrightarrow \quad (2\sin\theta - 1)(\sin\theta + 2) = 0.$$

As $\sin\theta \neq -2$, we get $\sin\theta = \frac{1}{2}$, so $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$. Then $x = \frac{2}{3}\cos\theta$ gives $x = \frac{1}{\sqrt{3}}$ or $x = -\frac{1}{\sqrt{3}}$ respectively, and in both cases $x^2 = \frac{1}{3} = \frac{2}{3}\sin\theta$, as required. The solutions are

$$(x,\theta) = \left(\tfrac{1}{\sqrt{3}}, \tfrac{\pi}{6}\right) \quad \text{and} \quad (x,\theta) = \left(-\tfrac{1}{\sqrt{3}}, \tfrac{5\pi}{6}\right).$$

Part (ii)

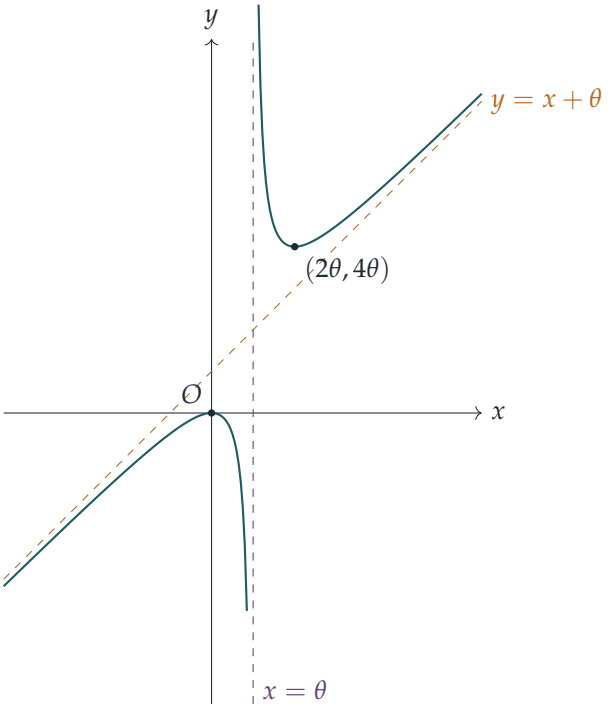
The sketch. Here $\theta > 0$. The curve $y = \dfrac{x^2}{x-\theta}$ passes through the origin (and meets the axes nowhere else), and has a vertical asymptote $x = \theta$. Dividing,

$$\frac{x^2}{x-\theta} = x + \theta + \frac{\theta^2}{x-\theta},$$

so $y = x + \theta$ is an oblique asymptote; the curve is above it for $x > \theta$ and below it for $x < \theta$. Differentiating with the quotient rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2x(x-\theta) - x^2}{(x-\theta)^2} = \frac{x(x-2\theta)}{(x-\theta)^2},$$

so the stationary points are $(0,0)$, a local maximum, and $(2\theta, 4\theta)$, a local minimum (the gradient changes sign from $+$ to $-$ at $x = 0$ and from $-$ to $+$ at $x = 2\theta$).



The deduction. If $x < \theta$, the numerator $x^2 \geq 0$ and the denominator is negative, so $y \leq 0$. If $x > \theta$, the branch has its lowest point at $(2\theta, 4\theta)$, so $y \geq 4\theta$. Algebraically, for $x > \theta$,

$$x^2 - 4\theta(x-\theta) = (x-2\theta)^2 \geq 0 \quad \Longrightarrow \quad \frac{x^2}{x-\theta} \geq 4\theta,$$

dividing by the positive quantity $x - \theta$. So the curve never takes values strictly between 0 and 4θ.

The trigonometric bound. The numerator satisfies $\sin^2\theta\cos^2 x \leq 1 \cdot 1 = 1$, and the denominator satisfies $1 + \cos^2\theta\sin^2 x \geq 1$. A number at most 1 divided by a number at least 1 (and positive) is at most 1:

$$\frac{\sin^2\theta\cos^2 x}{1+\cos^2\theta\sin^2 x} \leq 1.$$

Equality needs both $\sin^2\theta\cos^2 x = 1$ and $\cos^2\theta\sin^2 x = 0$. Note also that the fraction is ≥ 0, being a square over a positive number.

The final equation. Write $L = \dfrac{x^2}{4\theta(x-\theta)}$ and R for the right-hand side. Dividing the earlier result by $4\theta > 0$, either $L \leq 0$ or $L \geq 1$. But $0 \leq R \leq 1$.

Ask yourself Where can a number that is ≤ 0 or ≥ 1 meet a number in $[0,1]$?

Only at 0 or at 1. So we need $L = R = 0$ or $L = R = 1$.

- $L = 0$ forces $x = 0$, but then $R = \sin^2\theta > 0$ since $0 < \theta < \pi$. No solutions.
- $L = 1$ means $x^2 = 4\theta(x - \theta)$, i.e. $(x - 2\theta)^2 = 0$, so $x = 2\theta$. $R = 1$ needs $\sin^2\theta\cos^2 x = 1$, which forces $\sin^2\theta = 1$ and $\cos^2 x = 1$. With $0 < \theta < \pi$ this gives $\theta = \frac{\pi}{2}$, so $x = \pi$, and indeed $\cos^2\pi = 1$ and $\cos^2\theta\sin^2 x = 0$.

The only solution is $x = \pi$, $\theta = \frac{\pi}{2}$.

Checks

- Substitute $x = \frac{1}{\sqrt{3}}$, $\theta = \frac{\pi}{6}$ into the quartic: $9 \cdot \frac{1}{9} + (9 - 6) \cdot \frac{1}{3} - 12 \cdot \frac{1}{\sqrt{3}} \cdot \frac{\sqrt{3}}{2} + 4 = 1 + 1 - 6 + 4 = 0$.
- In the final equation with $x = \pi$, $\theta = \frac{\pi}{2}$: $L = \pi^2/(2\pi \cdot \frac{\pi}{2}) = 1$ and $R = 1 \cdot 1/(1 + 0) = 1$.
- A computer search over a fine grid of (x, θ) found no other near-solutions of either equation, apart from spurious ones creeping in at the excluded endpoints $\theta \to 0$ and $\theta \to \pi$ (where $\sin\theta \to 0$ and the bounds collapse).

Common slips

- Finding $\sin\theta = \frac{1}{2}$ and stopping: x must satisfy $x = \frac{2}{3}\cos\theta$, which fixes its *sign* for each θ, so $(\frac{1}{\sqrt{3}}, \frac{5\pi}{6})$ is not a solution.
- Forgetting the case $L = R = 0$ in the final part; it has to be ruled out, and it is ruled out only because $\sin\theta \neq 0$.
- Claiming the fraction is ≤ 1 "because $\sin^2$ and $\cos^2$ are at most 1" without saying why the denominator doesn't spoil it.

Beyond STEP

The trick of trapping an equation between a lower and an upper bound that meet at a single value is a staple of analysis: it is how one proves that equality holds in inequalities such as AM–GM or Cauchy–Schwarz, and how the squeeze theorem pins down limits. Learning to ask "when is this inequality tight?" is often the quickest route to a solution. For more in this spirit, see Abbott, *Understanding Analysis*, or the problem collection by Siklos, *Advanced Problems in Mathematics*.

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